

BRIDGING LEGAL EXPERTISE AND LLMS: A COOPERATIVE LOGICAL REASONING FRAMEWORK FOR SENTENCING RECOMMENDATION

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ABSTRACT

Large language models have achieved state-of-the-art performance in many natural language processing tasks. However, their logical reasoning capabilities remain insufficient to meet the rigorous standards required in the judicial domain. Legal knowledge is structurally complex, making it challenging for general LLMs to achieve a comprehensive and precise understanding. We propose Sentencing Recommendation Logical Reasoning, a novel framework that integrates key factual elements with computable mathematical logic to enhance the adaptability of LLMs to legal reasoning tasks. The SRLR framework translates legal domain knowledge into a circuit satisfiability problem, transforming it into computable mathematical logic. This approach helps LLMs avoid errors caused by faulty logical reasoning or incomplete knowledge of the legal domain. We conducted extensive experiments using a Traffic Accident Crime dataset, demonstrating that the SRLR framework not only produces stable and reliable outcomes but also significantly outperforms baseline models.

Index Terms— Large Language Models, Legal Element Extraction, Logical Reasoning, Sentencing Recommendation

1. INTRODUCTION

Large Language Models (LLMs) have revolutionized natural language processing, demonstrating remarkable capabilities in tasks ranging from text generation and question answering to complex multi-agent interactions [1, 2, 3]. However, their application in high-stakes, specialized domains such as law presents fundamental challenges that cannot be resolved by simply scaling model size or training data.

LLMs lack the deep domain expertise that legal professionals acquire through years of study. Legal knowledge is often encoded in dense, specialized texts, and the scarcity of high-quality domain-specific data makes it difficult to instill

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What is the content of Article 400 of the China's Criminal Law?



Article 400 of the Criminal Law of the People's Republic of China deals with the crime of dereliction of duty by military personnel. It states that military personnel who commit acts of negligence or fail to perform their duties properly, leading to significant consequences, are subject to criminal penalties. The penalties may include fixed-term imprisonment, criminal detention, or other forms of punishment, depending on the severity of the consequences caused by the negligence.

What is the content of Article 133 of the China's Criminal Law?



Article 133 of the Criminal Law of the People's Republic of China pertains to traffic accidents caused by violations of transportation regulations. It specifies that:
• Whoever violates transportation regulations, thereby causing major traffic accidents, resulting in serious injury or death of others, or serious damage to public or private property, shall be sentenced to fixed-term imprisonment of not more than three years or criminal detention.
• If the case is especially serious, the offender shall be sentenced to fixed-term imprisonment of not less than three years but not more than seven years.
This article is intended to punish individuals who, through negligence or reckless behavior while operating a vehicle or engaging in other forms of transportation, cause significant harm or loss.

Fig. 1: Two wrong results that directly ask GPT-4o. In the first dialogue, the researcher queries China's Criminal Law, which does not exist, but GPT-4o returns the wrong content. In the second dialogue, the researcher queries about TAC, GPT-4o fails to provide the full content of Article 133.

this expertise through standard pre-training or fine-tuning techniques. And LLMs' reasoning capabilities are not inherently designed to ensure logical soundness or strict adherence to formal rules [4, 5, 6]. The risk of hallucination [7] becomes particularly problematic in fields like law, where precision and verifiable reasoning are foundational.

These limitations are acutely evident in judicial applications, particularly in sentencing recommendations within China's legal system. An accurate sentencing decision requires both a precise understanding of case facts and the rigorous application of complex legal statutes [8, 9]. As shown in Figure 1, even state-of-the-art models like GPT-4o frequently fail to simultaneously interpret case narratives and follow the structured logic of legal reasoning. Simply prompting or fine-tuning an LLM to act as a "legal expert" remains insufficient and unreliable for real-world deployment.

To solve these problems, we propose the Sentencing Recommendations Logical Reasoning (SRLR) framework,

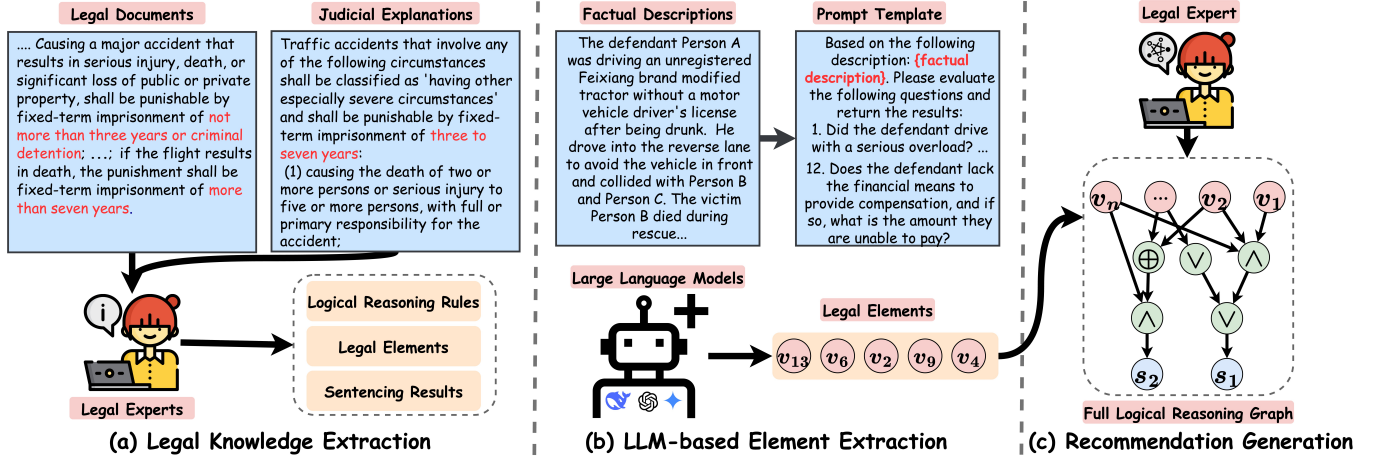


Fig. 2: Sentencing Recommendations Logical Reasoning Framework.

a novel approach that decouples unstructured text comprehension from formal legal reasoning. SRLR framework strategically leverages the strengths of both LLMs and symbolic reasoning systems. As illustrated in Figure 2, the SRLR framework operates in three phases: (a) **Legal Knowledge Extraction**: Legal experts first analyze judicial documents and legal explanations to distill logical reasoning rules, legal elements, and sentencing outcomes. This knowledge is then consolidated to construct a comprehensive logical reasoning graph; (b) **LLM-based Element Extraction**: We take the factual description of the case into a predefined prompt template. The LLM identifies the relevant legal elements present in the text; (c) **Recommendation Generation**: Finally, the extracted elements are fed into the logical reasoning graph, which traverses its paths to produce a specific sentencing recommendation. Our contributions are as follows:

- We propose SRLR, a novel framework that decouples natural language understanding from formal legal reasoning. It allows LLMs to focus on text processing while ensuring recommendations adhere to verifiable logical rules, effectively bridging the gap between LLM capabilities and the requirements of high-stakes domains.
- We introduce a new method to formalize complex legal knowledge. By representing legal statutes and their relationships as a **circuit satisfiability problem**, we transform the legal reasoning process into a solvable, transparent, and mathematically rigorous task using operators like “AND” ($\wedge$) and “OR” ($\vee$).
- We conduct extensive experiments on the “Traffic Accident Crime (TAC)” cause of action in China’s judicial system. Our results demonstrate that the SRLR framework significantly and consistently improves the accuracy of sentencing recommendations generated by various LLMs, validating the effectiveness of our approach.

2. RELATED WORK

Legal domain knowledge in LLMs. Legal domain knowledge is important in China’s civil law. Existing studies apply LLMs to the judicial domain in different ways, such as fine-tuning the LLM [10, 11], using the RAG (Retrieval Augmented Generation) techniques to optimize the output of LLMs [12, 13]. In sentencing recommendation and judge prediction, Deng [8] improved the performance of LLMs in judge prediction by fine-tuning LLM with legal documents.

Mathematical logic and logical reasoning. Mathematical logic has been widely utilized across various fields within computer science, including areas such as logical reasoning, knowledge graph, and LLMs [14, 15]. Pan [16] explored the performance of LLMs to translate natural language problems into mathematical logic. Sun [17] incorporates mathematical logic into LLMs through memory-based approaches, further improving their logical reasoning and inference capabilities.

3. METHODOLOGY

3.1. Preliminaries

We build our sentencing recommendations graph $G = (V, E, S)$ using mathematical logic and Circuit Satisfiability Problem [18]. We regard the legal elements as the proposition set $V = \{v_1, v_2, \dots, v_n\}$, where n is the number of legal elements. Based on the statute documents, judicial interpretations, and other sources, legal experts compile legal elements v . The logical relationships (logical operators) between legal elements are represented as $E = \{\wedge, \vee, \neg, \dots, \oplus\}$. We define sentencing recommendations as the result of compound propositions $S = \{s_1, \dots, s_m\}$, where m is the number of the standard sentencing recommendations in the statute document.

Let D be the factual description dataset, and $d \in D$ a factual description. We define f_V as the function of the LLM

extracting the legal elements in d . The propositions of legal elements expressed in the factual description d are

$$v_{i,d} = f_V(d, p_e), 1 \leq i \leq n,$$

where $v_{i,d}$ is the expression of v_i in d , and p_e is the prompt for retrieving the legal elements in the factual description. We use $V_d = \{v_{1,d}, \dots, v_{n,d}\}$ to represent all elements that exist in the factual description d . The sentencing recommendation of d can be determined as

$$s_d = f_E(v_{1,d}, \dots, v_{n,d}), 1 \leq i \leq n,$$

where the function f_E represents the logical reasoning function for sentencing recommendations.

3.2. Legal Elements Definition

We use TAC to illustrate the definition of legal elements and the building of the logical reasoning graph. We ask legal experts to extract and summarize the professional legal knowledge in statute documents to obtain the legal elements and sentencing reasoning rules required for TAC. Then, we compile a set of legal elements that influence sentencing recommendations based on the accumulated professional knowledge of the experts, standard statute documents, and judicial interpretations: Serious overload driving (SOD, v_1); Knowingly unlicensed driving or scrapped motor vehicle (KUD, v_2); Driving a motor vehicle without driving qualification (DMV, v_3); Driving a motor vehicle after drinking or taking drugs (DMVAD, v_4); Driving motor vehicle knowing that the safety device is incomplete or the safety mechanism is malfunctioning (DMVIM, v_5); Number of seriously injured (NSI) includes seriously injured one or two persons (v_6), three or four persons (v_7), more than five persons (v_8); Number of death (ND) that include one person die (v_9), two persons die (v_{10}), three to five persons die (v_{11}), more than six persons die (v_{12}); Responsibility determination (RD) that include full responsibility (v_{13}), primary responsibility (v_{14}), equal responsibility (v_{15}); Flee (Flee, v_{16}); The death was caused by fleeing (DF, v_{17}); No ability to compensate (NoCom) that include 300,000 to 600,000 (v_{18}), more than 600,000 (v_{19}). The proposition set, based on the legal elements of TAC, is represented as $V = \{v_1, \dots, v_{19}\}$.

3.3. Logical Reasoning Graph

Following Article 133 of the Criminal Law of China and the judicial interpretation of TAC, we categorized the sentencing recommendations for TAC into three ranges: less than three years, three to seven years, and more than seven years. The result of the compound proposition set is $S = \{s_1, s_2, s_3\}$. Figure 3 shows the TAC's fully mathematical logical reasoning path and the circuit satisfiability graph.

Many existing studies require AI to output a sentence-length accurate to the month. However, these studies actually do not conform to general law requirements and judicial

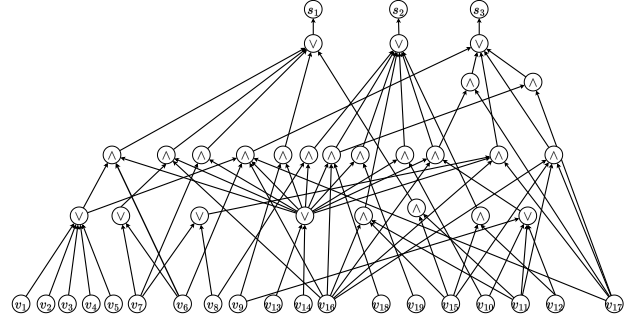


Fig. 3: Full TAC sentencing recommendations logical reasoning graph.

decision-making. According to the sentencing guidelines and provisions issued by the Supreme People's Court of China, there are three steps for judges to decide on a sentence.¹ First, they need to decide which category the sentence shall fall in based upon essential criminal facts. Second, within the range of the category, they need to decide on a rough sentence based on criminal facts, plus essential circumstantial facts. Third, this rough sentence shall be further adjusted to produce a final sentence, considering other circumstantial facts.

3.4. Legal Elements Extracted by LLM

We utilize LLMs to extract legal elements from d , and compile the proposition set, V , through accumulated legal knowledge, statute documents, and case-handling experience of the legal experts. We do not require LLMs to have specialized legal knowledge to assist in extracting legal elements. Each legal element has been organized into knowledge of a general domain, such as the number of fatalities or injuries. Based on V , we employ LLMs to extract each legal element from the factual description d , ultimately obtaining V_d :

Step 1: Constructing the prompt p_e . We do not know which legal elements are present in the factual description d . Therefore, we need to inquire about all possible legal elements. As shown in phase b of Figure 2, we put the factual description d into LLMs using a prompt.

Step 2: Organizing the propositions set of legal elements V_d . We use LLMs to extract the legal elements from the factual description d and to get the result V_d . We then input the legal elements from V_d into the logical reasoning graph (Figure 3) to get the sentencing recommendations.

4. EXPERIMENTS

4.1. Setting and Evaluation Metrics

Dataset. We construct our dataset with the judgment documents of TAC published in the Judicial Decision Database²

¹<https://www.court.gov.cn/shenpan/xiangqing/6622.html>

²<https://wenshu.court.gov.cn/>

Table 1: Accuracy and f1-score results of different LLMs.

Model	Approach	SRLR		Question		Legal Question	
		Acc	Wei.F1	Acc	Wei.F1	Acc	Wei.F1
GPT-4o		92.20	92.81	65.20	69.58	81.40	84.76
Deepseek-R1		92.17	92.68	91.03	91.37	92.13	92.37
Claude-3-opus		92.10	92.63	82.00	80.29	84.47	84.47
Gemini-1.5-pro		92.03	92.43	57.40	62.24	78.10	80.42
Qwen2.5-max		91.53	92.22	84.40	84.76	88.57	89.78
Doubao-pro-32k		91.47	92.27	85.23	85.90	87.27	88.28
Deepseek-chat		91.10	92.36	40.73	43.01	67.33	71.30
GPT-4		89.57	91.13	47.87	52.20	38.17	41.91
Llama3.1-70b		89.20	90.63	49.60	55.57	58.67	64.56
Gpt-4o-mini		80.03	84.63	34.37	35.94	38.20	40.28

after 2021. In 2021, the Supreme People’s Court of China released the latest judicial interpretation of sentencing recommendations³. The dataset contains 2468 judgment documents with sentences of less than three years, 526 with sentences of three to seven years, and 6 with sentences of more than seven years.

Ground Truth. To construct the ground truth for our experiments, we employed six legal experts to annotate the judgment documents of TAC. They mapped the final sentencing outcome of each case to one of three sentencing ranges.

Evaluation Metrics. Our data set is imbalanced. We employed the weighted average evaluation metrics and general evaluation metrics to evaluate our final experimental results.

4.2. Implementation Details

Baseline. Based on the statute documents and judicial interpretations, we employ two approaches as our baseline experiments using large language models:

- **Question.** We input the factual descriptions from the judgment documents into an LLM without providing legal knowledge. The LLM returns a sentence term and maps it to one of three sentencing ranges.
- **Question with Legal Knowledge (Legal Question).** We compile and summarize the statute documents about TAC, judicial interpretations, and other legal knowledge that are necessary for sentencing recommendations. These legal documents are fed into the LLM as the prior legal domain knowledge required for legal sentencing.

Table 2: Precision and recall results of different LLMs.

Model	Approach	SRLR		Question		Legal Question	
		Wei.P	Wei.R	Wei.P	Wei.R	Wei.P	Wei.R
GPT-4o		93.65	92.20	84.76	65.20	88.71	81.40
Deepseek-R1		93.57	92.17	92.09	91.03	92.89	92.13
Claude-3-opus		93.48	92.10	79.47	82.00	84.50	84.47
Gemini-1.5-pro		93.20	92.03	80.43	57.40	85.40	78.10
Qwen2.5-max		93.37	91.47	85.22	84.40	92.11	88.57
Doubao-pro-32k		93.37	91.47	87.13	85.23	90.46	87.27
Deepseek-chat		93.92	91.10	85.02	40.73	87.10	67.33
GPT-4		92.93	89.57	81.74	47.87	85.37	38.17
Llama3.1-70b		92.49	89.20	79.03	49.60	80.50	58.67
Gpt-4o-mini		92.24	80.03	84.36	34.37	85.19	38.20

³<https://www.court.gov.cn/zixun/xiangqing/312301.html>

4.3. Experiment Results

As shown in Table 1, the SRLR framework outperforms both the “Question” and “Legal Question” approaches in terms of the F1-score evaluation metric. For different LLMs, the SRLR framework can significantly improve the accuracy of sentencing recommendations. The SRLR framework could enhance the performance of LLMs to generate sentencing recommendations. The worst-performing GPT-4o-mini model also achieves an accuracy of 80%. In contrast, the accuracy of the “Question” and “Legal Question”, for GPT-4o-mini, are 34% and 38%. By comparing the accuracy between “Question” and “Legal Question”, we can conclude that legal domain knowledge is beneficial in the context of civil law.

In China’s judicial domain, LLMs such as Deepseek-R1, Doubao, and Qwen have superior performance in the “Question” and “Legal Question”. Our guess is that the training data of these models may already contain legal knowledge in China’s judicial domain. The SRLR framework does not require additional legal knowledge corpora to fine-tune LLMs, leading to a low training cost.

As shown in Table 2, applications in the judicial domain require more rigorous and stable outcomes. Even the least performant GPT-4o-mini demonstrated significant improvement in weighted precision and weighted recall. The performance of our SRLR framework are more stable and outperforms all other approaches in all evaluation metrics. Moreover, the experimental results indicate that providing legal knowledge to LLMs improves their performance. When we accurately input the required legal knowledge into the LLMs, they generate sentencing recommendations based on the provided legal knowledge.

5. CONCLUSION

In this paper, we introduce the SRLR framework, which enhances the performance of LLMs in sentencing recommendations. By integrating legal domain knowledge and logical reasoning, the SRLR framework improves evaluation metrics, addressing the challenges LLMs face with legal reasoning.

An interesting future research direction is the automated generation of logical reasoning graphs from existing legal documents and judicial interpretations. By leveraging RAG techniques and the textual processing capabilities of LLMs, it would be possible to analyze and process the intrinsic logical relationships among legal elements, ultimately constructing comprehensive and structured reasoning graphs.

6. ACKNOWLEDGMENTS

This work is supported in part by Important Project of National Social Science Fund of China (No.25AFX004, titled “The Computational Expression of Law”) and Fapu Tech Ltd. Co.

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